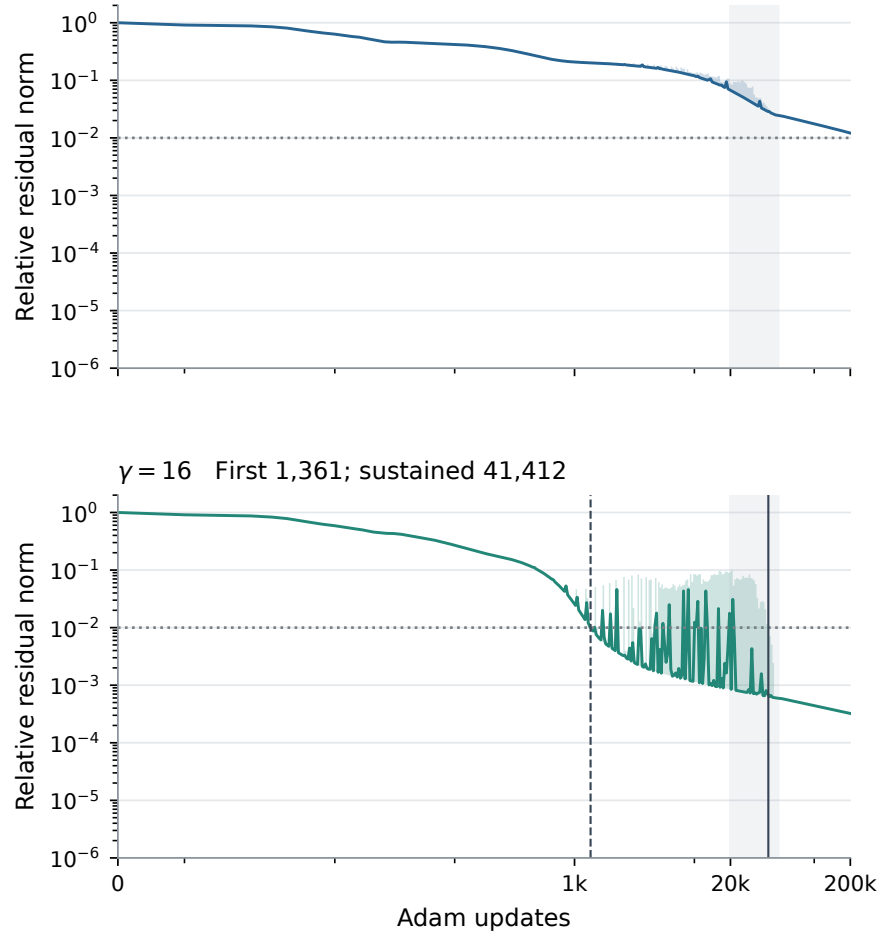


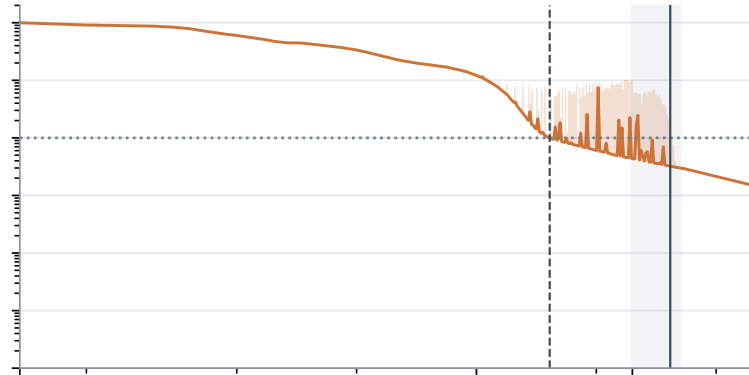
First crossing can precede stable acquisition

Sine mixture · common Adam setting · shaded vertical window: learning-rate decay (20k-50k)

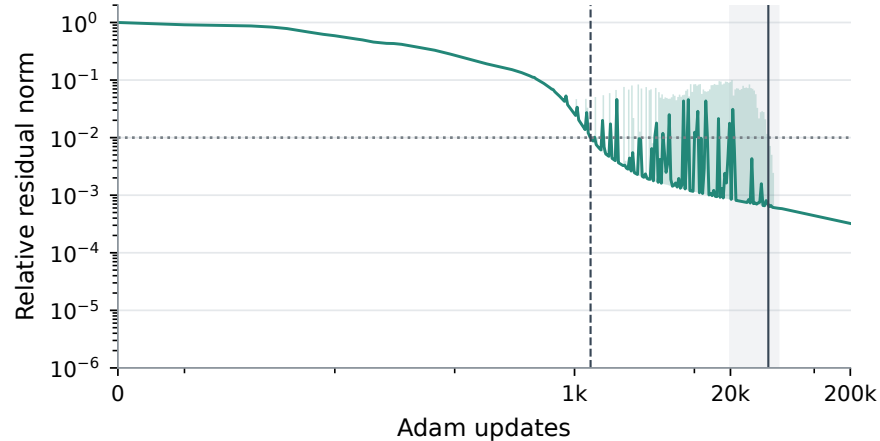
$\gamma = 8$ First >200,000; sustained >200,000



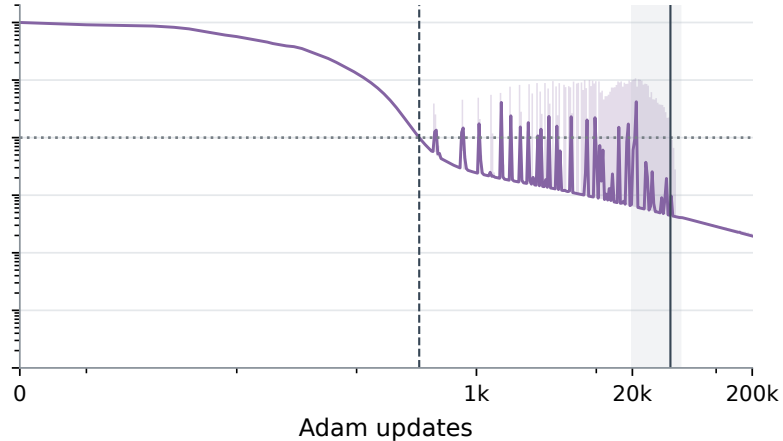
$\gamma = 12$ First 4,084; sustained 41,397



$\gamma = 16$ First 1,361; sustained 41,412



$\gamma = 64$ First 333; sustained 41,539



■ All-update min-max envelope — Sampled actual residual